

Name
Class
R. No.

• Mathematics • IIT JEE / CBSE
• Test 2 (July 17 , 2010) • Sets
• Maximum Marks 45 • Time 30 minutes

Instructions

1. The test consists of two sections. Section I carries 20 Marks , section II carries 25 Marks.
2. The questions carry marks for partially correct solutions also.

Section I

Q1: There are 200 individuals with a skin disorder , 120 have been exposed to the chemical C_1 , 50 to the chemical C_2 and 30 to both the chemicals C_1 and C_2 . Find the number of individuals exposed to

- i) Chemical C_1 but not chemical C_2 .
- ii) Chemical C_2 but not chemical C_1 .
- iii) Chemical C_1 or chemical C_2 .

15 Marks

Q2: In a committee , 50 people speak French , 20 speak Spanish and 10 speak both Spanish and French.
How many speak at least one of these two languages?

5 Marks

Section II

Q3: For any sets A and B, show that

$$P(A \cap B) = P(A) \cap P(B)$$

10 Marks

Q4. A market research group conducted a survey of 1000 consumers and reported that 720 consumers like product A and 450 like product B, what is the least number that must have liked both products?

5 Marks

Q5: In a survey it was found that 21 people liked product A, 26 liked product B and 29 liked product C. If 14 people liked products A and B , 12 people liked products C and A , 14 people liked products B and C and 8 liked all the three products. Find how many liked product C only.

10 Marks
